Research Methods — Module 04: Cognition — Study Questions

1. Why is behavioral task design critical for testing Active Inference? What is the dissociation principle?
2. How do you manipulate precision in a behavioral task? Give two examples.
3. How do you manipulate volatility? What behavioral measures reflect volatility estimation?
4. Why are eye movements considered direct evidence of active inference?
5. What does fixation duration at a particular location reveal about uncertainty?
6. How does scanpath analysis reveal the brain's hypothesis-testing strategy?
7. What does Active Inference predict about visual search termination?
8. How does pupil dilation relate to prediction error (surprise)?
9. What is the relationship between tonic pupil size and decision uncertainty?
10. How does the locus coeruleus-norepinephrine system connect to precision modulation?
11. How does probabilistic reversal learning dissociate learning rate from volatility estimation?
12. What is the "sampling problem"? How does Active Inference provide a stopping rule?
13. What determines planning depth in Active Inference? How might you measure it?
14. What is a MouseLab paradigm? How does it reveal information sampling strategies?
15. What control conditions are necessary for a volatility manipulation?
16. How does computational phenotyping address individual differences in behavioral tasks?
17. What confounds threaten the interpretation of pupil dilation as "surprise"?
18. How do practice effects affect computational parameter estimates?
19. Design a behavioral task dissociating epistemic value from pragmatic value.
20. Critically evaluate: Can behavioral data alone distinguish Active Inference from reinforcement learning accounts?